

Practice Quiz: The Testing Agent

Part A: Multiple Choice

1. The creator-evaluator transition is difficult because: A) Building prototypes requires different tools than testing them B) The inventor's strong priors about the design create precision-weighted bias toward confirming observations C) Inventors are not trained in statistical methods D) Testing requires more people than building
2. In Active Inference, an inventor with high sensory precision during testing will: A) Ignore all test data and rely on intuition B) Give strong weight to incoming test data and update beliefs quickly C) Build higher-fidelity prototypes D) Recruit fewer external testers
3. Beta testers are valuable as "distributed sensors" primarily because: A) They work for free B) They each bring a unique generative model, generating diverse prediction errors C) They are more technically skilled than the inventor D) They provide positive feedback that helps with marketing
4. An adversarial tester's goal is to: A) Confirm that the prototype works as designed B) Provide encouraging feedback to the inventor C) Maximize surprise by finding conditions under which the prototype fails D) Reduce the cost of testing by finding quick shortcuts
5. Confirmation bias in prototype testing occurs when: A) The inventor tests too many prototypes B) The inventor's priors are so strong that they dominate the posterior, preventing belief updating from test data C) Too many testers agree that the prototype works D) The test equipment is faulty
6. Pre-registration in testing involves: A) Registering the invention with the patent office before testing B) Committing to specific hypotheses, protocols, and success criteria before testing begins C) Registering all testers in a database D) Pre-ordering materials for the next prototype before finishing the current test

7. The Segway case study illustrates: A) The importance of gyroscopic stabilization B) How the creator's observation model can miss social and practical prediction errors that external users immediately detect C) That consumer electronics always fail D) That adversarial testing is unnecessary for transportation devices

Part B: Short Answer

1. Explain why watching a user struggle with your prototype without intervening is both difficult and valuable. Use Active Inference concepts (prediction error, priors, active inference) in your answer.
2. You are developing a mobile app for elderly users. Design a testing agent strategy that includes at least three types of agents, specifying what each agent's generative model prioritizes and what unique prediction errors each is likely to generate.
3. Two members of your invention team observe the same user test. One concludes the prototype succeeded; the other concludes it failed. Using the concept of observation models and precision weighting, explain how this is possible and propose a method for resolving the disagreement.